NLP Assignment 3 Group 6

Cristian Alin Aron (S5236401)
Enzo Castelo Branco Biondi (S6579434)
Bogdan-Gabriel Ivancu (S5160901)

April 2026

1 Introduction

Assignments 1 and 2 established a strong pipeline for AG News 4-way classification. Assignment 1 achieved a test Macro-F1 of 0.9145 with TF-IDF and LinearSVC, and Assignment 2 improved this to 0.9180 with a BiLSTM encoder trained on word embeddings. This assignment fine-tunes a pretrained Transformer encoder (DistilBERT) on the same task under identical data conditions, then evaluates robustness via two complementary slice analyses: performance stratified by input length, and label-noise sensitivity using reduced training sets (25%, 50%, and 100% of training data). The goal is to understand not only whether DistilBERT outperforms prior models, but also where it remains fragile.

2 Dataset and Split Details

All data conditions are identical to Assignments 1 and 2 to allow direct comparison. The AG News dataset (HuggingFace: `sh0416/ag_news`) provides a predefined split of 120,000 training instances and 7,600 test instances across four classes: World, Sports, Business, and Sci/Tech (1,900 instances per class in the test set). A validation set was carved out from the training data using a 90/10 split with random seed 36, yielding 108,000 training and 12,000 validation instances. The test set was evaluated exactly once, after model selection on the validation set. Text input combines the title and description fields, lower-cased and stripped of backslashes, consistent with Assignments 1 and 2.

3 Transformer Model

3.1 Architecture

DistilBERT-base-uncased (Sanh et al., 2019) was used as the pretrained encoder. DistilBERT is a distilled version of BERT that retains 97% of BERT's perfor-

mance on GLUE benchmarks while being 40% smaller and 60% faster. The pretrained model was loaded via HuggingFace Transformers with a classification head (a linear layer mapping the `[CLS]` token representation to four output logits) freshly initialised for the AG News downstream task. Input sequences were tokenised with DistilBERT's WordPiece tokenizer and truncated or padded to a maximum length of 128 tokens.

3.2 Training Hyperparameters

All hyperparameters are reported in Table 1.

Table 1: DistilBERT fine-tuning hyperparameters.

Hyperparameter	**Value**
Model checkpoint	distilbert-base-uncased
Max sequence length	128 tokens
Batch size (train / eval)	16 / 16
Learning rate	2×10^{-5}
Weight decay	0.01
Optimizer	AdamW (HuggingFace default)
Number of epochs	4
Early stopping patience	5 epochs (val Macro-F1)
Random seed	36
Hardware	Tesla T4 GPU (Google Colab)

3.3 Training Progress

Table 2 shows per-epoch training and validation metrics for the full training run (108,000 instances). Training loss decreases steadily across all four epochs. Validation loss begins to rise after epoch 2, indicating the onset of overfitting, while validation Macro-F1 plateaus. The best checkpoint (epoch 2, Macro-F1 = 0.9416) was selected via the `load_best_model_at_end` flag. Figure 1 shows the corresponding learning curves.

Table 2: Per-epoch metrics for DistilBERT (full training set).

Epoch	**Train Loss**	**Val Loss**	**Val Accuracy**	**Val Macro-F1**
1	0.2341	0.2107	0.9331	0.9335
2	0.1428	0.1979	0.9412	0.9416
3	0.0975	0.2604	0.9403	0.9407
4	0.0638	0.2968	0.9405	0.9409

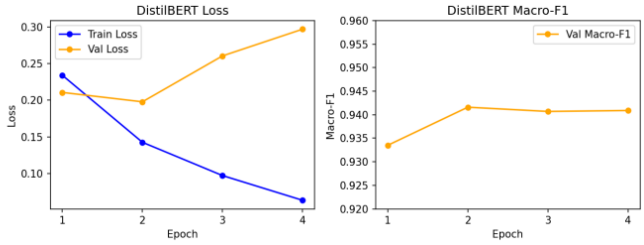

Figure 1: DistilBERT training and validation loss (left) and validation Macro-F1 (right) — full training set.

4 Test Results

The best checkpoint was evaluated once on the holdout test set (7,600 instances). Table 3 compares DistilBERT against the best models from Assignments 1 and 2.

Table 3: Comparison of best models across assignments (test set).

Model	**Test Accuracy**	**Test Macro-F1**
LinearSVC + TF-IDF (A1)	0.9145	0.9145
BiLSTM + Dropout 0.3 (A2)	0.9182	0.9180
DistilBERT fine-tuned (A3)	0.9441	0.9442

DistilBERT achieves the highest scores across both metrics, improving test Macro-F1 by +2.62 points over the A2 BiLSTM and +2.97 points over the A1 LinearSVC. The improvement is consistent with the expectation that pretrained contextual representations provide richer features than bag-of-words (TF-IDF) or randomly initialised word embeddings.

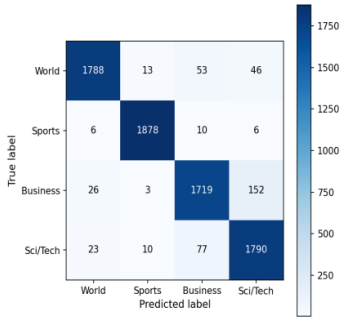

Figure 2: DistilBERT confusion matrix on the test set (7,600 instances).

Table 4 shows the confusion matrix on the test set. The Business ↔ Sci/Tech axis remains the dominant confusion source, with 152 Business instances predicted as Sci/Tech and 77 Sci/Tech instances predicted as Business. This mirrors the pattern observed in Assignments 1 and 2, suggesting a structural overlap in vocabulary for technology company articles. Sports remains the cleanest class with only 22 errors total. World shows modest confusion with both Business (53) and Sci/Tech (46), driven by articles about economic policy and geopolitical technology topics.

Table 4: DistilBERT confusion matrix on the test set (7,600 instances).

	World	**Sports**	**Business**	**Sci/Tech**
World	1788	13	53	46
Sports	6	1878	10	6
Business	26	3	1719	152
Sci/Tech	23	10	77	1790

5 Robustness / Slice Evaluation

5.1 Length Bucket Evaluation

To assess whether input length affects model quality, the test set was partitioned into four word-count bins: 0–20, 21–40, 41–60, and 61+ words. Accuracy and Macro-F1 were computed independently for each bin. Results are reported in Table 5.

Table 5: DistilBERT performance by input length bucket.

Length Bucket	**N**	**Accuracy**	**Macro-F1**
0–20 words	206	0.9029	0.8775
21–40 words	4,712	0.9416	0.9411
41–60 words	2,508	0.9533	0.9533
61+ words	174	0.9253	0.8964

The model performs best on medium-length inputs (41–60 words, F1 = 0.9533) and degrades at both extremes. Very short inputs (0–20 words, F1 = 0.8775) suffer because they lack sufficient context to distinguish ambiguous topics — a headline such as "Market falls" could belong to World, Business, or Sci/Tech. Long inputs (61+ words, F1 = 0.8964) are relatively few (174 instances) and show a non-trivial drop, possibly because they exceed the model's 128-token window and relevant class-indicative content gets truncated. The 21–40 word bucket dominates the test set (62% of instances) and achieves performance close to the overall test score, confirming the model is well-calibrated for the typical article length in AG News.

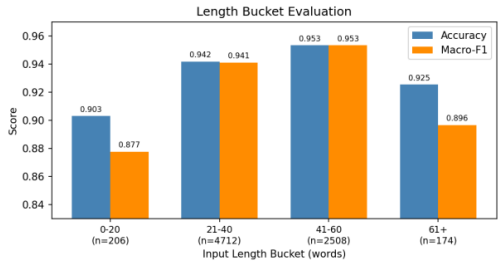

Figure 3: Accuracy and Macro-F1 per input length bucket.

5.2 Label-Noise Sensitivity (Training Size Reduction)

To assess sensitivity to reduced supervision — a proxy for label-noise robustness — DistilBERT was retrained from scratch using 25% (27,000), 50% (54,000), and 100% (108,000) of the training set, with all other hyperparameters held constant. The same validation and test splits were used throughout. Table 6 reports test-set performance across the three conditions, and Table 7 and Table 8 show per-epoch logs for the 50% and 25% runs respectively.

Table 6: DistilBERT test performance at different training set sizes.

Training Size	Train Instances	Test Accuracy	Test Macro-F1
25%	27,000	0.9337	0.9337
50%	54,000	0.9395	0.9396
100%	108,000	0.9441	0.9442

Table 7: Per-epoch metrics for DistilBERT (50% training set).

Epoch	Train Loss	Val Loss	Val Accuracy	Val Macro-F1
1	0.2663	0.2328	0.9284	0.9289
2	0.1568	0.2340	0.9344	0.9349
3	0.1028	0.2890	0.9314	0.9319
4	0.0643	0.3296	0.9307	0.9311

Table 8: Per-epoch metrics for DistilBERT (25% training set).

Epoch	Train Loss	Val Loss	Val Accuracy	Val Macro-F1
1	0.3024	0.2477	0.9186	0.9190
2	0.1705	0.2593	0.9238	0.9244
3	0.1063	0.3201	0.9233	0.9236
4	0.0688	0.3419	0.9248	0.9253

Performance degrades gracefully as training data is reduced: Macro-F1 drops from 0.9442 (100%) to 0.9396 (50%) to 0.9337 (25%), a total decline of only 1.05 points. This stability reflects the strength of pretrained representations: DistilBERT has already encoded rich general language knowledge, so it requires relatively few labelled examples to adapt to the downstream task. Notably, even at 25% training data, DistilBERT (F1 = 0.9337) outperforms the full-data BiLSTM from Assignment 2 (F1 = 0.9180), suggesting that the pretrained features generalise efficiently. The per-epoch logs for 50% and 25% show slightly higher validation losses compared to 100%, consistent with less signal for fine-tuning, but early stopping still selects the best epoch.

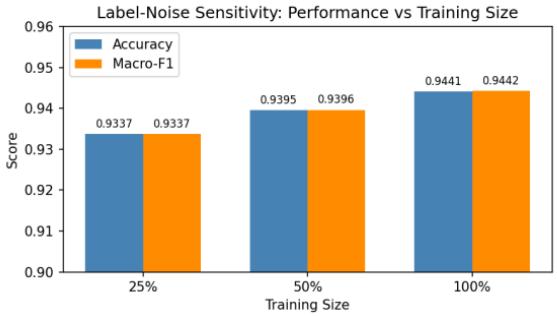

Figure 4: Test accuracy and Macro-F1 as a function of training size.

6 Error Analysis

The model made 425 errors on the 7,600 test instances (overall error rate: 5.6%), compared to 622 for the A2 BiLSTM and 648 for the A1 LinearSVC. This represents a reduction of 197 errors relative to A2, confirming that pretrained contextual representations substantially improve classification quality across all confusion pairs.

Per-class error breakdown

The dominant confusion axis remains Business ↔ Sci/Tech, accounting for 229 of the 425 errors: 152 Business instances were predicted as Sci/Tech, and 77 Sci/Tech instances were predicted as Business. This asymmetry is informative — the model is more likely to over-predict Sci/Tech for Business articles than the reverse, suggesting that technology company names (IBM, Intel, Yahoo) act as strong Sci/Tech anchors even when the article's primary framing is financial or operational.

World is the second most error-prone class, contributing 112 misclassifications in total. The most common confusions are World → Business (53 errors) and World

→ Sci/Tech (46 errors), both driven by articles about macroeconomic policy, oil markets, and geopolitical technology topics where topic-boundary vocabulary is genuinely ambiguous. Compared to A2, DistilBERT makes substantially fewer World → Sports errors (13 vs. 38), suggesting that the pretrained model has learned to distinguish geopolitical event coverage from sports reporting more reliably.

Sports remains the cleanest class with only 22 total errors, consistent with the class having the most distinctive vocabulary in the dataset. The small number of Sports → World misclassifications typically involve sports scandal articles that blend institutional language (IOC, inquisitors, investigations) with athlete-centred content.

Failure modes

Three recurring failure modes emerge from inspection of misclassified examples. First, **technology company ambiguity**: articles about companies like IBM, Yahoo, and Intel generate both Business and Sci/Tech signals simultaneously. When such articles focus on workforce decisions, pricing strategy, or competitive positioning, the company name still biases the model toward Sci/Tech — a form of entity-level anchoring that contextual representations partially but not fully resolve. Second, **topic-indicative keyword override**: certain words ("fraud", "IOC", "oil prices") carry strong class associations that can override the article's broader topic. An article about electoral fraud in Venezuela is misclassified as Sci/Tech because fraud co-occurs frequently with cybersecurity content in training data. Third, **input brevity**: very short articles lack the redundant contextual cues that help the model triangulate the correct class, which is consistent with the length-bucket analysis showing the lowest F1 (0.8775) for the 0–20 word bin. Many of the hardest examples are short headlines with no description, where even a human reader might struggle to assign a unique label.

7 Limitations

Truncation at 128 tokens. AG News articles combined with descriptions can exceed 128 tokens. The length-bucket analysis shows a non-trivial performance drop for the 61+ word bin (F1 = 0.8964). Longer-context models (e.g., Longformer) or an increased `max_length` could address this, at additional computational cost.

Business ↔ Sci/Tech boundary. Despite outperforming all prior models, DistilBERT still makes 229 errors on this single confusion axis. Richer supervision signals — such as source-domain metadata or multi-label annotations — may be required to resolve this structural ambiguity in the dataset.

Single fine-tuning run. All reported results come from a single run per configuration with no variance estimate. DistilBERT fine-tuning is known to

exhibit high variance across seeds for small datasets; although AG News is large enough to reduce this risk, a multi-run average would strengthen the conclusions.

8 Reproducibility Notes

All experiments were run on Google Colab with a Tesla T4 GPU (Python 3.12, PyTorch, HuggingFace Transformers). The full source code and notebook are available at:

Table 9: Reproducibility safeguards.

Safeguard	**Detail**
Random seed	`SEED = 36` used in all splits, sampling, and model init
Data splits	Identical to A1/A2: 108k train, 12k val, 7.6k test
Test evaluation	Test set evaluated exactly once after val-based model selection
Dataset source	HuggingFace: `sh0416/ag_news` (official predefined split)
Environment	`requirements.txt` with exact package versions in repository
Checkpoints	All three model checkpoints (25%, 50%, 100%) saved and available

Appendix: Model Card - DistilBERT AG News Classifier

- **Model Details:** DistilBERT-base-uncased fine-tuned for 4-way sequence classification. 6 layers, 768 hidden size, 12 attention heads, 66M parameters.
- **Intended Use:** Automated categorization of English news headlines and descriptions into World, Sports, Business, and Sci/Tech.
- **Factors:** Evaluation performed on the AG News test set. Performance may vary on non-English text or specialized domains (e.g., medical news).
- **Metrics:** Primary: Accuracy (94.41%), Secondary: Macro-F1 (0.9442%).
- **Training Data:** 108,000 instances from AG News.
- **Limitations:** Performance drops on texts shorter than 20 words. High confusion between Business and Sci/Tech for technology company reports.
- **Ethical Considerations:** The model may reflect biases present in the 2004-era AG News dataset (e.g., gender distribution in sports).

.